UPLOAD SLOT 03 OF 12

Concept Plans and Spatial Organization Diagrams

How the park is organised — the crescent, the radial alleys, and the rooms between them.

CONTENTS

Al Safa 2 Park — Falaj Al Safa

Mohamed Wasim · Individual Applicant

Generated 2026-08-03 · reproducible: python tools/build_submission_pdfs.py

Design Narrative & Concept

Falaj Al Safa — an arc of shade for a park that is currently too hot to use

Of the 4,402 daylight hours in a Dubai year, only **44.5%** are comfortable to stand in on the open site today. This proposal raises that to **64.6%** — a gain of 20.1 percentage points, or about 884 additional usable hours a year.

Upload slot 01 — Design Narrative & Concept
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The problem, stated as a number

A neighbourhood park nobody can stand in is not a park. Al Safa 2's difficulty is not layout, planting or maintenance — it is thermal. The site is flat and largely exposed, at 25°N, where the summer sun is nearly overhead and the winter sun is low enough to slide underneath most shade.

The analysis behind this submission reconstructs an 8,760-hour year for the site from 39 years of National Centre of Meteorology monthly normals, verified back against those normals to within 0.39 °C. Across the 4,402 daylight hours in that year, the open site is comfortable for 44.5% of them. That is the number the design exists to move.

2. The concept

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

Element	What it is	Description
Al Hilal	the Crescent Canopy	18 m gridshell at 4.5 m over a 7 m walk, with a 3 m southern louvre
Al Falaj	the water channel	0.9 m wide, on the canopy's drip line so it is shaded all day and evaporates less
Al Nakhil	the Oasis Basin	a sunken palm court held in the crescent's concave side
Al Sikkak	the alleys	radial — each one a radius of the same arc
Al Madar	the perimeter loop	a tree-shaded running and walking circuit
Al Kathib	the dune berm	planted earth against the roads — noise, glare and heat

The six named elements. All are generated from a single arc definition in src/plan.py.

3. Why an arc — the part that was solved, not styled

A straight canopy presents one orientation. When a sun angle defeats it, it defeats the whole length at once, and the walk has no shade anywhere along it. An arc changes heading continuously, so some segment is always angled well.

The depth of the bow was not chosen for appearance. It was swept against the 8,760-hour solar model at a fixed section, over all 4,402 daylight hours of the year. The criterion was the final column — the hours in which the route offers nowhere at all to stand.

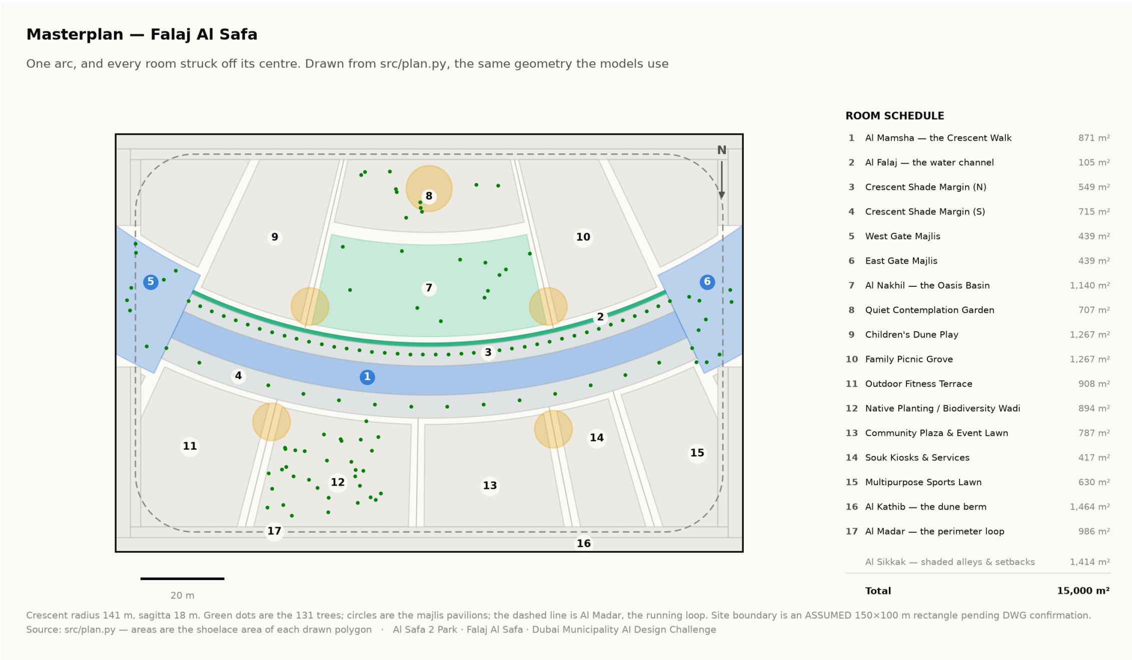
Plan form	Mean cover	Worst month	Hours with no shade anywhere
Straight east–west bar	87.4%	68.7%	330
Sine meander (superseded)	85.0%	73.1%	63
Arc, sagitta 10 m	87.1%	70.2%	116
Arc, sagitta 14 m	86.6%	71.3%	62
Arc, sagitta 18 m	85.9%	72.1%	52
Arc, sagitta 22 m	84.9%	72.3%	61
Closed elliptical loop	79.1%	69.8%	89

Adopted row highlighted. Source: the sweep recorded in src/config.py, measured against the same solar model used throughout this submission.

Read the first column carefully. The straight bar has the *highest* mean coverage of anything tested — an east–west canopy is close to the optimum orientation for 25°N, and curving it costs about a point. The crescent is not claimed to shade more ground on average. It is claimed to remove the hours in which the route offers nowhere at all to stand, and it removes six sevenths of them. That is the trade, stated in the direction that is not flattering.

The closed loop loses on every measure, for the same reason the deep bows do: it forces half its length to run north–south, and a canopy over a north–south route can only work when the sun is low in the east or west. The loop survives in the scheme as Al Madar, an *unshaded* running circuit — because that is what a circuit is actually for.

The arc bows convex south. The solar model is indifferent to that sign, so the choice was spatial: bowing south puts the structure between the sun and the hollow it wraps, making the concave side the park's cool pocket instead of a south-facing bowl. Every room people are asked to linger in sits in that hollow.



Masterplan. Every area is the shoelace area of the drawn polygon, which is why the schedule closes on 15,000 m² without anyone reconciling a spreadsheet. Technical drawing — to scale.

4. What the design achieves

Measure	Today	As designed
Comfortable daylight hours	44.5%	**64.6%**
Peak heat index	56.8 °C	**48.7 °C**
Mean heat-index reduction under canopy	—	**7.13 °C**
Crescent Walk shaded (canopy and louvre)	—	**87.3%**
Site-wide mean shade	—	34.1%
Trees	—	131

Site-wide mean shade is modest at 34.1%, and that is a position, not an oversight. This scheme makes a few places genuinely excellent rather than the whole site marginally better. A park with even shade everywhere and no continuously comfortable route would score higher on that one statistic and be worse to use.

5. Two corrections this submission makes to itself

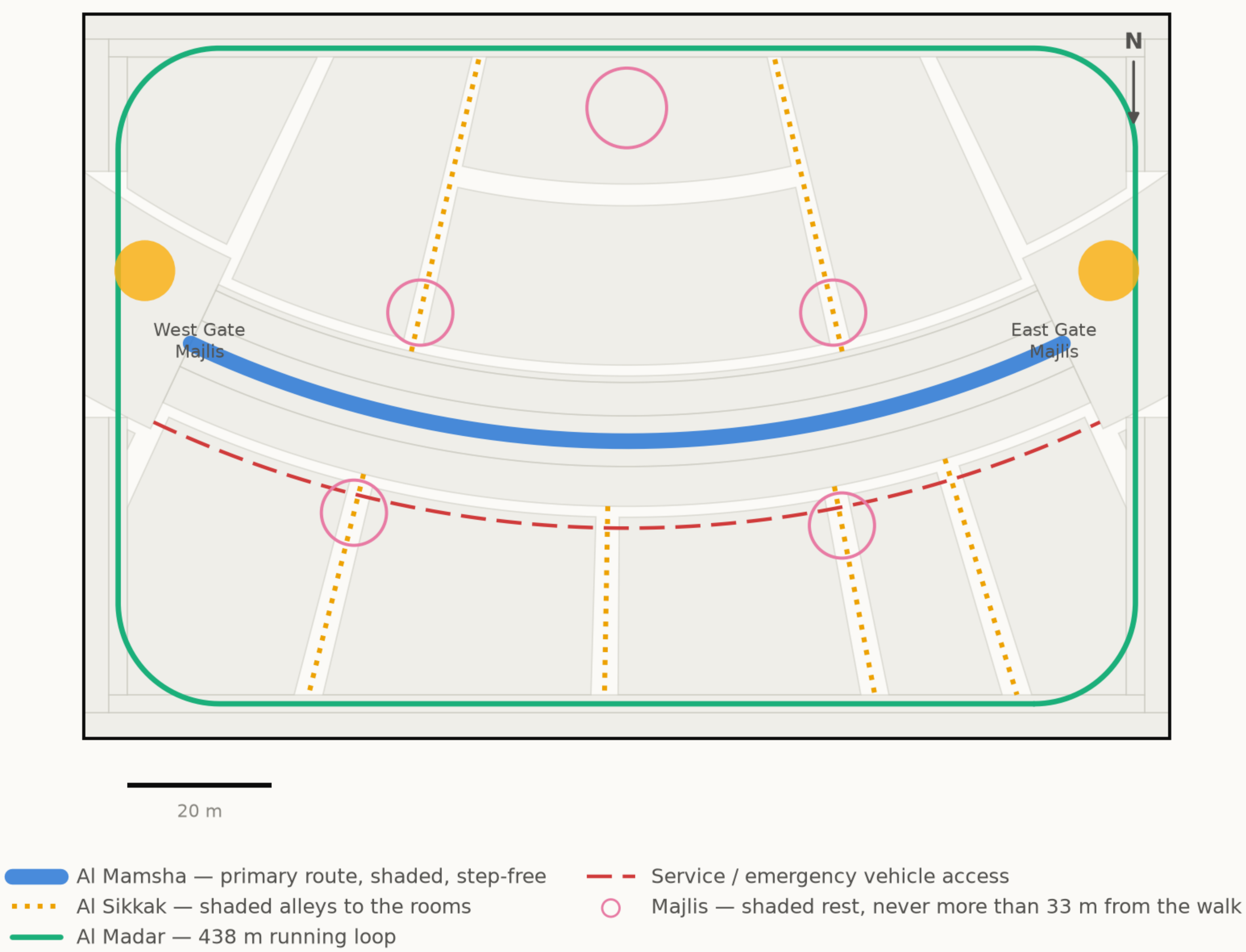
A withdrawn shade claim. An earlier version claimed 99.2% annual shade on a flat 9 m canopy over a 9 m walkway. It does not survive a geometric check — when the sun is low and to the south, the shadow slides clean off the path — and it was withdrawn. The section was then re-solved: a narrower walk, a wider overhang, a lower plane and a deep southern louvre. It now measures 87.3%, and that is a number that can be checked.

Three withdrawn images. Three visuals presented invented data as measurement: a thermal comfort analysis attributed to CFD when no CFD was run; a satellite NDVI analysis whose raster was noise; and a parametric canopy described as solar-optimised when no objective was ever defined. All three were withdrawn on the same principle as the 99.2%: anything that cannot survive being checked should not be in front of a juror.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Circulation & accessibility

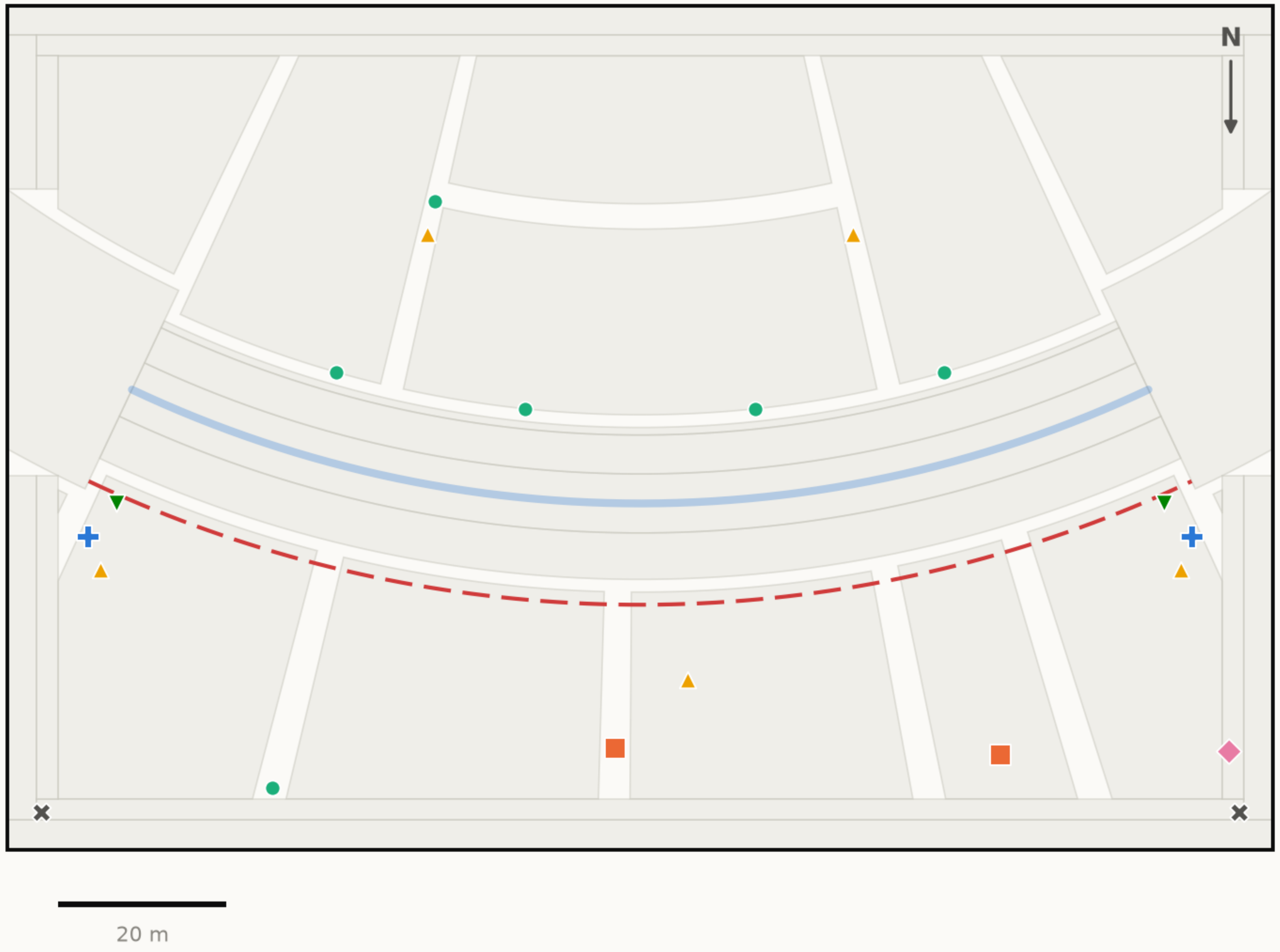
One shaded primary route, radial alleys off it, and a running loop around the whole



Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Commercial & service facilities

Where the park earns its running costs, and where it is serviced



- Service / emergency vehicle access
- Commercial — souk kiosks and café (2)
- Public restrooms (2)
- Drinking fountain (6)
- Waste & recycling station (5)
- Service & maintenance store (1)
- Bicycle parking (2)
- Drop-off / pick-up bay (2)

20 facilities. Commercial floor sits on the convex south face (417 m2, 2.8% of the site), facing the plaza and the event lawn so that evening trade and programming reinforce each other — and away from the concave hollow, which is kept quiet. Every facility is reachable from the dashed service route without a vehicle entering the walk.

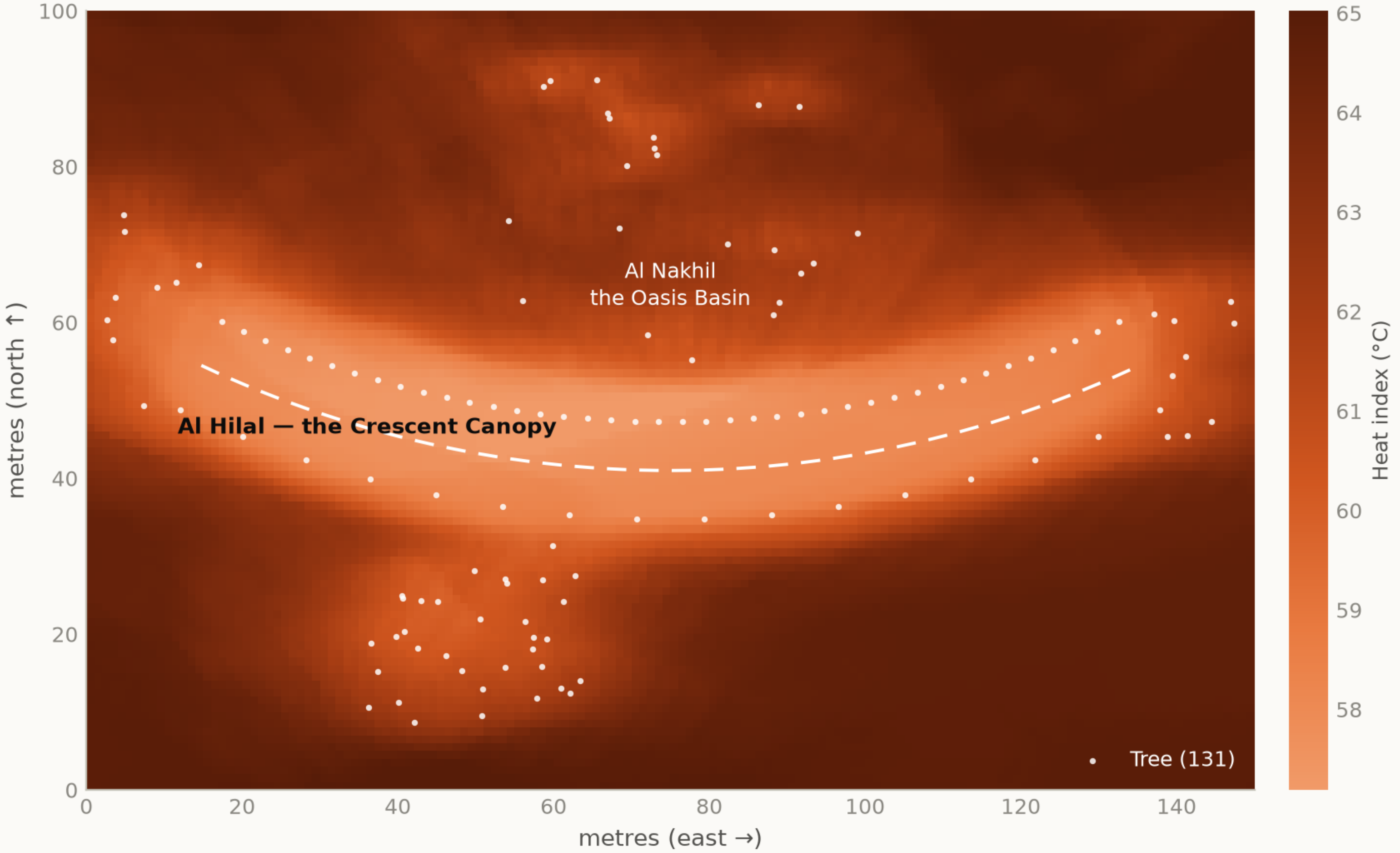
Source: src/plan.py — facility positions are arc coordinates, so they move with the crescent · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig04 site comfort map

Analysis output — computed from project data.

Predicted summer comfort across the site

July afternoon heat index per square metre — the shade strategy, tested

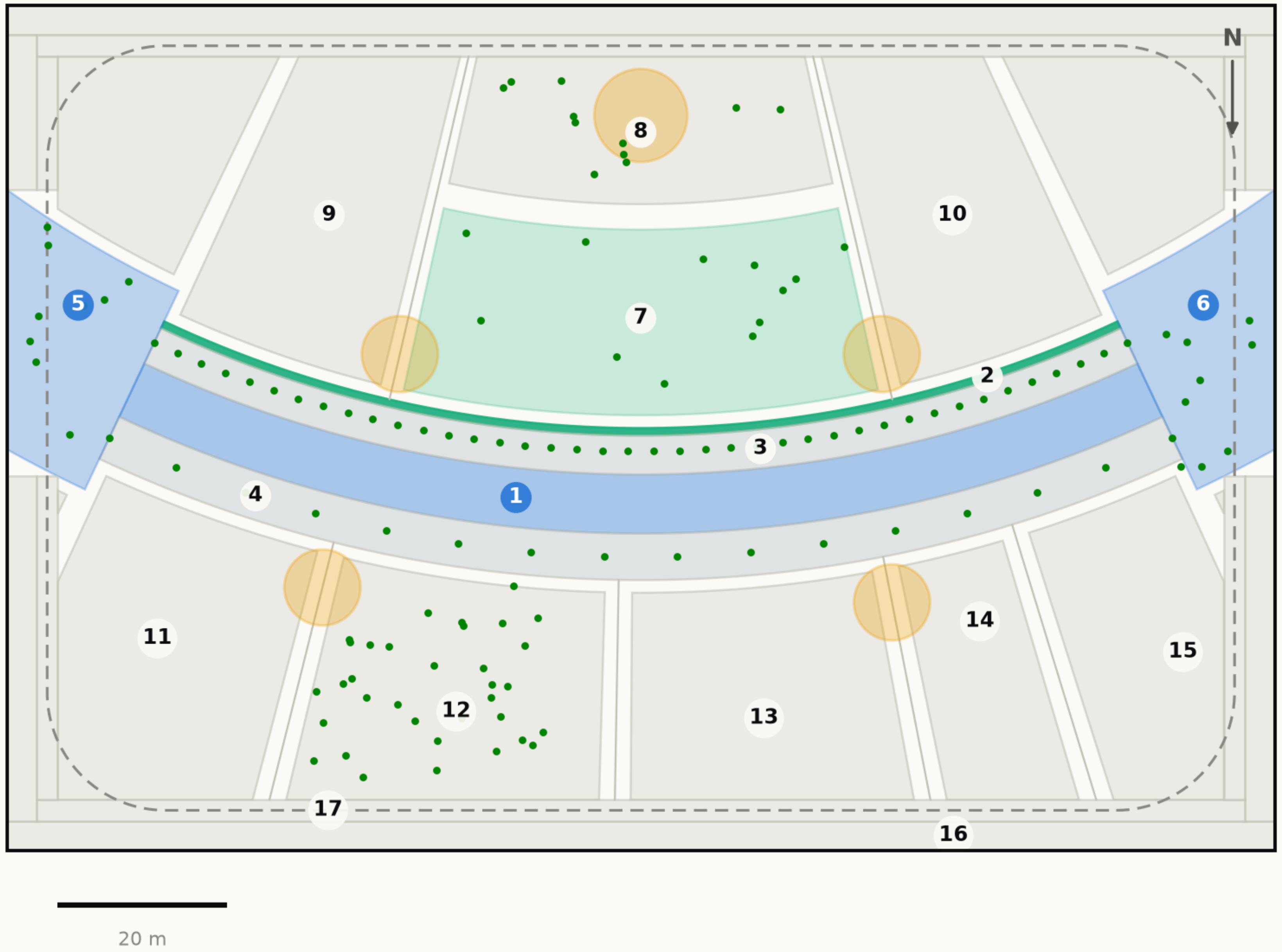


Lighter is cooler. The crescent reads as the coolest continuous route across the site, and its concave side is measurably cooler than its convex one — which is why the rooms people are asked to linger in are all placed there. White dots are the 131 trees.

Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; NCM Dubai climate normals 1977-2015 (39-year record) · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²
	Total	**15,000 m²**

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge